CSCI 788 CAPSTONE · MILESTONE 1

Grounded Multi-Agent RAG for Healthcare Workflow Automation

Checking Medicare coverage questions against public policy, with a citation behind every finding.

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Coverage rules are long, and a wrong answer is costly

01 Long rules

Three CMS manual chapters on home health, ambulance and inpatient care total about 75,000 words.

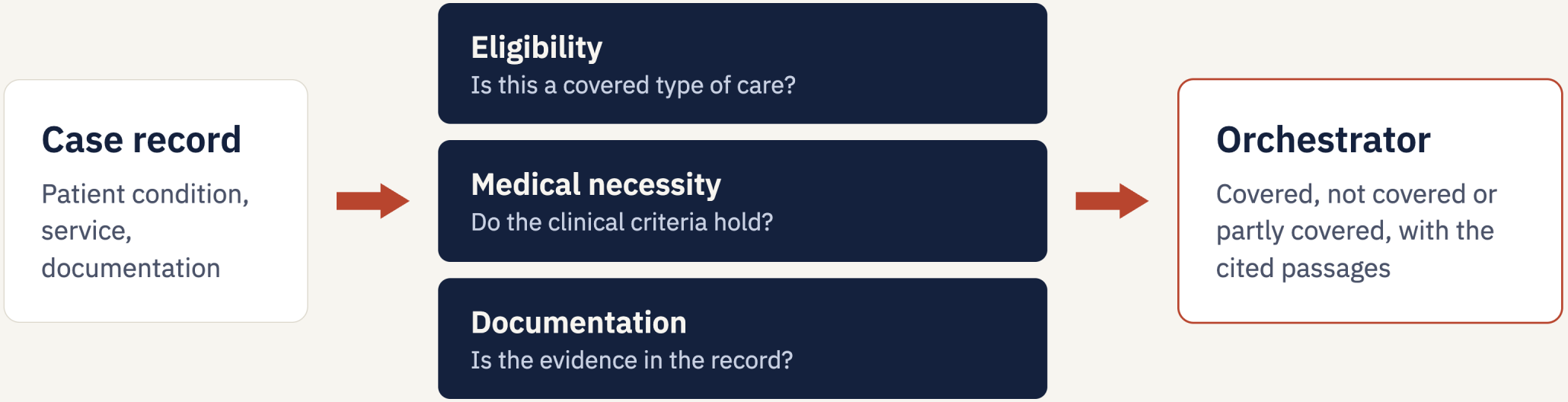
02 Manual review

A reviewer finds the right rule, matches it to the patient's record, then decides.

03 Invented rules

A plain chatbot can state a rule that does not exist. In healthcare that is unacceptable.

One case, three specialist agents, one cited decision



Each agent retrieves the policy passage it relies on and cites it. A single-agent RAG system is the baseline for comparison.

All data is public.

POLICIES

Three CMS manual chapters, unmodified

- Medicare Benefit Policy Manual ch. 7, Home Health (2023 revision)
- Ch. 10, Ambulance (2009 revision)
- Ch. 1, Inpatient Hospital Services (2021 revision)

TEST CASES

22 real Medicare Appeals Council decisions

- 10 home health, 7 ambulance, 5 inpatient and rehab
- Outcomes: 11 favorable, 8 unfavorable, 3 partial
- Decided by Medicare judges, who cite the policy they applied

Our only annotation: the patient facts are copied word for word from each decision, and the outcome and cited sections are recorded separately, so the outcome never enters the system's input.

Decisions: HHS Departmental Appeals Board, public decisions page. Policies: cms.gov. Raw files and text are in the repository.

Method selected: chunk, retrieve, check, cite

Chunking

463 citable passages from the three chapters, each labeled with chapter and section number.

Retrieval baseline

TF-IDF with cosine similarity. The query is the case facts only, never the outcome.

Answer key

The manual sections the Council itself cited. Not our wording, so the score is not circular.

Split

Development and test cases, fixed seed, balanced by topic. Tuning on development only.

Findings will be MET, NOT MET or UNKNOWN per criterion, each with a citation. Milestone 2 adds LLM-written findings and dense-embedding retrieval under the same citation rule.

Initial pipeline: working end to end

463
policy passages indexed

22
real decisions run as cases

4
LLM roles: three agents and an orchestrator

1
single-agent baseline, same cases

Flow: ingest, retrieve, three agents, orchestrator, cited decision. Every prompt and response is saved, the outcome never appears in a prompt, and everything is committed to git. Model for this run: Claude Sonnet 5.5.

Retrieval: does the search find the judges' section?

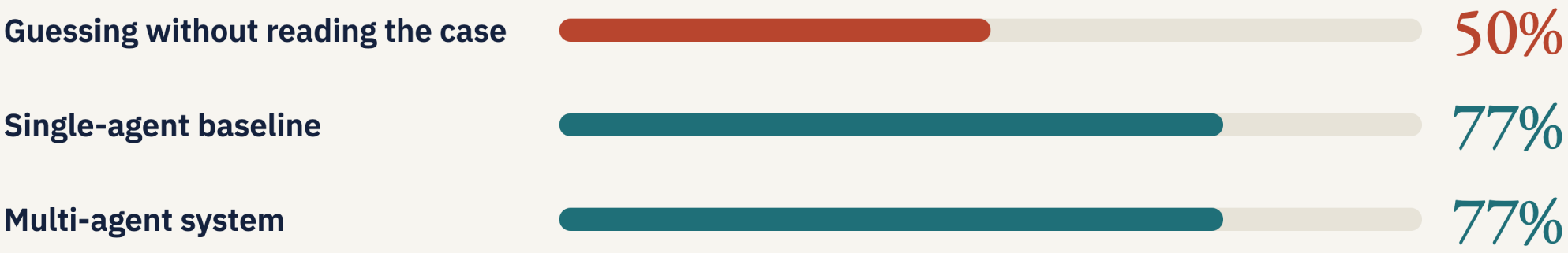
For each case the system pulls up the best-matching passages. A hit means the section the judges cited is among them.

	IN THE TOP 5 PASSAGES	IN THE TOP 10 PASSAGES
Exact section The very section the judges cited	32%	49%
Exact or related section Same section, or its parent or child	61%	83%

20 cases scored. Simple baseline on purpose; dense embeddings come in Milestone 2. Output: results/retrieval_v2.json

Decision results: multi-agent and single-agent tie

How often the system's answer (covered, not covered, partly covered) matches the real Council outcome, over 22 cases.



DEVELOPMENT, 12 CASES

Multi-agent **75%**

Single-agent **67%**

TEST, 10 CASES

Multi-agent **80%**

Single-agent **90%**

TAKEAWAY

They disagree on only 2 of 22 cases.

On this set it is a tie.

Citation script check: 97% multi-agent, 100% single-agent. Model calls per case: 4 versus 1. Output: results/milestone1_decisions.json

Limits of the data and the pipeline, and the fix

Policy versions differ

Decisions cite 2009 to 2012 policy; two chapters are newer. 24 cited sections no longer exist. Fix: archived manual versions.

Appeals, not requests

These are appeals after payment, not requests before service. Same reasoning, different stage. Fix: state the scope.

Small set

22 cases, 10 for testing. A one-case swing is 10 points. Fix: about 90 more candidate decisions are public.

AI in the loop

Facts and outcomes were extracted with AI help and only spot-checked. The deciding model may also have seen these public decisions. Fix: second reader, newer cases.

No multi-agent gain yet

17 of 22 for both systems. Fix: more cases, repeated runs, and a human check of the findings.

Weak retrieval

TF-IDF misses half the exact sections at top 10. Fix: dense embeddings, tuned on development only.

Honest limits are part of the result.

Measures: what exists now and what Milestone 2 adds

Metric	Definition	Status
Retrieval recall	Share of Council-cited sections found in the top results	Measured: 49% exact, 83% parent/child at top 10
Decision agreement	Match with the Council outcome: covered, not covered, partly covered	Measured: 77% multi-agent, 77% single-agent
Groundedness	Share of findings supported by their cited passage	Script check measured; human check in Milestone 2
Hallucination rate	Share of findings citing text that is not in the policy or case	Script check: 3% of multi-agent findings; human check next
Latency	Seconds per case, multi-agent versus single-agent	Calls per case measured (4 versus 1); seconds in Milestone 2

Milestone 2 reports every metric on the 10 held-out test cases, plus more cases, with the single-agent baseline alongside.

Next steps toward Milestone 2

1. Human in the loop: a reviewer confirms the case facts, checks that each citation supports its claim, and decides on low-confidence cases.
2. Run the agents in a live loop to measure latency and how stable the decisions are.
3. Add more public decisions and archived manual versions; second reader on a sample.
4. Improve retrieval with dense embeddings, tuned on the development cases only.
5. Test whether the multi-agent design beats the baseline on harder cases.

Input requested: how the review step should work, and whether a local open model is acceptable.